

THE DISTANCE-REDSHIFT RELATION FOR UNIVERSES WITH NO INTERGALACTIC MEDIUM

C. C. DYER* AND R. C. ROEDER†

Kitt Peak National Observatory,‡ Tucson, Arizona

Received 1972 April 19

ABSTRACT

The distance-redshift relation is derived for model universes in which there is negligible intergalactic matter and in which the line of sight to a distant object does not pass close to intervening galaxies. When fitted to observations, this relation yields a higher value of q_0 than does a homogeneous model.

We shall consider here a simplified model of the Universe in which all the matter is concentrated into objects such as galaxies. For a beam of light propagating far from any of these objects, the optics very closely approximates that of a beam propagating down a small empty tube in an otherwise homogeneous universe. For this case, the optical scalar equations (Sachs 1961) in the form used by Kantowski (1969) reduce to

$$(A^{1/2})'' = 0, \quad (1)$$

where A is the cross-sectional area of the beam and a prime denotes the derivative $\equiv d/d\tau$, τ being the affine parameter along the beam. In particular, consider the conical beam with vertex at the observer and base at the extragalactic object being observed. The angular diameter distance (in units of c/H_0) is then related to A by

$$D_{<} \propto A^{1/2}, \quad (2)$$

so that $D_{<}$ satisfies

$$D_{<}'' = 0, \quad (3)$$

and hence we have

$$D_{<} = B\tau + C. \quad (4)$$

For a Robertson-Walker model (McVittie 1965a) τ must satisfy

$$d\tau \propto Rdt \quad (5)$$

(Schrödinger 1956), so, using the standard expression for dR/dt , we have (up to a multiplicative constant which we may take as 1)

$$d\tau/dz = (1+z)^{-3}(2\sigma_0(1+z) + (\sigma_0 - q_0)(1+z)^{-2} + 1 + q_0 - 3\sigma_0)^{-1/2}, \quad (6)$$

where σ_0 and q_0 are the density and acceleration parameters (McVittie 1965b). For convenience, we take $\tau = 0$ at $z = 0$ and from equation (6) we then have $\tau \approx z$ for very small z . But $D_{<} \approx z$ for very small z and hence in equation (4) $B = 1$, $C = 0$.

For the rest of the calculation, we shall assume that $\sigma_0 = q_0$; i.e., that $\Lambda = 0$. Then

* Permanent address: David Dunlap Observatory, Richmond Hill, Ontario.

† On leave from Scarborough College and David Dunlap Observatory.

‡ Operated by the Association of Universities for Research in Astronomy, Inc., under contract with the National Science Foundation.

we have

$$D_{<} = \int_0^z \frac{dz}{(1+z)^3(1+2q_0z)^{1/2}}. \quad (7)$$

This integral can be reduced to the integral giving ω , the standard angular radial coordinate of the Robertson-Walker models. After some manipulation, we obtain, for $q_0 < 0.5$,

$$D_{<} = \frac{3q_0^2}{(1-2q_0)^{5/2}} \left\{ \sinh^{-1} \left(\frac{1-2q_0}{2q_0} \right)^{1/2} - \sinh^{-1} \left[\frac{1-2q_0}{2q_0(1+z)} \right]^{1/2} \right\} \\ - \frac{3q_0}{2(1-2q_0)^2} \left[1 - \frac{(1+2q_0z)^{1/2}}{1+z} \right] + \frac{1}{2(1-2q_0)} \left[1 - \frac{(1+2q_0z)^{1/2}}{(1+z)^2} \right]; \quad (8a)$$

for $q_0 = 0.5$,

$$D_{<} = \frac{2}{5} [1 - (1+z)^{-5/2}]; \quad (8b)$$

and for $q_0 > 0.5$,

$$D_{<} = \frac{3q_0^2}{(2q_0-1)^{5/2}} \left\{ \sin^{-1} \left(\frac{2q_0-1}{2q_0} \right)^{1/2} - \sin^{-1} \left[\frac{2q_0-1}{2q_0(1+z)} \right]^{1/2} \right\} \\ - \frac{3q_0}{2(2q_0-1)^2} \left[1 - \frac{(1+2q_0z)^{1/2}}{1+z} \right] - \frac{1}{2(2q_0-1)} \left[1 - \frac{(1+2q_0z)^{1/2}}{(1+z)^2} \right]. \quad (8c)$$

Equation (8b) was first presented, in slightly different form, by Zel'dovich (1964); equations (8a) and (8c) have not, as far as we can tell, been presented before.

Since $dD_{<}/dz = d\tau/dz$ has the form given by equation (6), it goes to zero only for infinite z . Hence, while $D_{<}$ has a maximum in the homogeneous case (as can be seen from Mattig's 1958 relation), $D_{<}$ has no maximum at finite z in this model but approaches an upper limit as z tends to infinity. Thus, the apparent size of a standard object tends to a finite limit as z tends to infinity.

Once we have obtained the $D_{<}(z)$ relation, the luminosity distance D_L (units of c/H_0) follows directly from a result due originally to Etherington (1933), namely,

$$D_L = (1+z)^2 D_{<}. \quad (9)$$

This model predicts a dimming effect relative to the homogeneous model generally used; for a given set of observations the value of q_0 derived with this model should be somewhat larger than that derived from the homogeneous model. As an illustration, we have used McVittie's (1972) reconstituted magnitude-redshift data (but using Oke's 1971 value of $z = 0.38$ for 0024 + 1654) and have found the least-squares best-fit value of q_0 for the inhomogeneous model and the homogeneous model, assuming in both cases, as before, that $\sigma_0 = q_0$. For the homogeneous model, we find $q_0 = 1.40$; and for the model considered here, we find $q_0 = 1.82$, showing the expected increase in q_0 .

The basic assumptions of this model are that there is little or no intergalactic matter and that we see distant galaxies along lines of sight that do not pass too close to intervening galaxies. In figure 1, the lower curve shows the dimming, in magnitudes, of the model considered here relative to the homogeneous model. The upper curve is the dimming, relative to the homogeneous model, calculated numerically for the corresponding Swiss cheese model (Kantowski 1969) with no intergalactic matter and assuming that the beam of observation passes no closer than 2 kpc to the center of a galaxy like our own. The probability of a closer approach is comfortably small for redshifts in the range of interest.

It can be seen that the dimming is appreciable in the relevant range of z , but that the

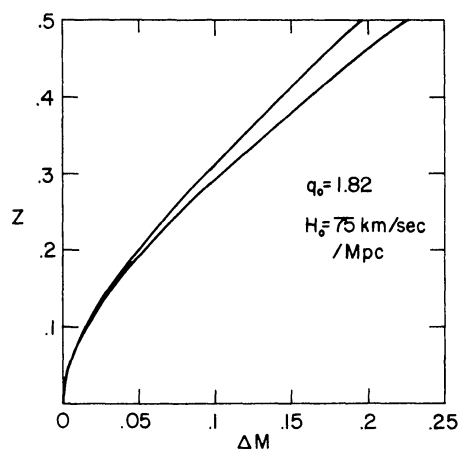


FIG. 1.—The dimming, relative to the homogeneous model, assuming that the beam passes far from any intervening galaxies (*lower curve*) and assuming that the beam passes no closer than 2 kpc to the center of galaxies similar to our own (*upper curve*).

difference between the model considered here and the detailed numerical integration is small compared with the total correction. Hence, there would seem to be some justification for using the simple relations (8a), (8b), and (8c) obtained here for comparison with the observational data.

Further detailed numerical integrations have been done for Swiss cheese models, with $\Lambda = 0$ and $\Lambda \neq 0$, and will be presented elsewhere. These calculations show that the mass deficiency in the beam is in general much more important than the gravitational lens effect for reasonable deflectors.

We acknowledge financial support from the National Research Council of Canada (C. C. D.) and Kitt Peak National Observatory (R. C. R.).

REFERENCES

- Etherington, I. M. H. 1933, *Phil. Mag.*, **15**, 761.
 Kantowski, R. 1969, *Ap. J.*, **155**, 89.
 McVittie, G. C. 1965a, *General Relativity and Cosmology* (Urbana: University of Illinois Press), equation (8.215).
 ———. 1965b, *ibid.*, pp. 149, 150.
 ———. 1972, *M.N.R.A.S.*, **155**, 425.
 Mattig, W. 1958, *Astr. Nachr.*, **284**, 109.
 Oke, J. B. 1971, *Ap. J.*, **170**, 193.
 Sachs, R. K. 1961, *Proc. Roy. Soc. London, A*, **264**, 309.
 Schrödinger, E. 1956, *Expanding Universes* (Cambridge: Cambridge University Press), chap. 2.
 Zel'dovich, Ya. B. 1964, *Soviet Astr.—AJ.*, **8**, 13.